What Is a Taxi-Out Time? Data Floors, Silent Substitutions and the Limits of Prediction in the PRC Data Challenge 2026

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Abstract

The PRC Data Challenge 2026 asks participants to predict taxi-out time for departures at ten major European airports. We report a study of what is predictable in that dataset and what is not. Our central finding is that the target variable is not always a taxi-out time: about 4.8 % of departures have their off-block timestamp silently replaced by the scheduled time, and a smaller family carries a date-rollover defect. Twenty-six rows out of 344,227 carry 58 % of the validation squared error. We show that one of these corrupted families is exactly reconstructible from columns that remain visible in the ranking file, worth 7.95 s of root mean squared error on the public leaderboard, while the other is explicable after the fact but has no usable identifying signal among the provided fields and signatures we tested. We further show that the classic congestion predictors of the taxi-time literature are screened off by the observed Network Manager off-block time, but that features built from the departures that *follow* the target flight are not, which reflects the difference between forecasting and retrospective reconstruction. Finally, we observe a 2.9 s range across three seeds of one fixed configuration and use it to mark several earlier single-seed conclusions as unresolved. We argue that a substantial share of our residual error is associated with the definitional gap between two off-block sources; we do not claim that the remaining gap to the leading models is irreducible.

1 Introduction

Taxi-out time, the interval between off-block and takeoff, is a standard efficiency indicator in air traffic management. It drives fuel burn and associated emissions on the ground, it is a core metric of Airport Collaborative Decision Making, and its predictability conditions departure metering schemes. The literature on its prediction is mature: reported root mean squared errors fall in the range of 75 s to 90 s on carefully measured physical taxiing [2], with queue size at pushback consistently identified as the dominant predictor since [1].

The PRC Data Challenge 2026, organised by the EUROCONTROL Performance Review Unit together with the OpenSky Network, provides twelve months of 2025 movement data at ten European airports and asks for the taxi-out time of departures in January and July 2026, scored by root mean squared error. The provided data pair each airport movement record with the corresponding Network Manager flight record where a match exists.

This paper is not a winning-method report. Our final public score, 319.21 s, places us in the lower half of the leaderboard, whose leader stands at 241.75 s. The contribution we claim is of a different nature. Working on a real operational dataset with an evaluation signal attached to it, we were able to measure things that practitioners encounter daily and rarely publish: how a target variable degrades when it is assembled from two information systems

with different definitions, how much of a competition metric can rest on a handful of defective records, which literature results survive a change of prediction setting, and how sensitive small model comparisons can be to training randomness.

We organise the paper around four claims.

1. The target is not always a taxi time, and the two corrupted families it contains behave completely differently with respect to prediction: one is exactly recomputable from provided columns; for the other, none of the provided fields or tested signatures yields a usable identifier (Section 3).

2. A substantial component of the error floor observed in our pipeline is a data floor. Once the corrupted rows are set aside, the residual remains strongly associated with the disagreement between the airport off-block and the Network Manager off-block, whose per-airport standard deviation lies between 250 s and 500 s (Section 5). This does not establish a universal lower bound for other models.

3. Congestion features are screened off by the observed Network Manager off-block, but only in the causal direction. Features built from subsequent takeoffs carry information that the literature's pushback-time features do not, because the challenge is a reconstruction task rather than a forecasting task (Section 6).

4. In one fixed configuration, three random seeds span 2.9 s. This limited diagnostic is not a variance estimate, but it is enough to make smaller single-seed effects unresolved, including several of our own earlier comparisons (Section 7).

The evidence comes from three distinct channels. Structural diagnostics and rule discovery use the 2025 training data; model comparisons use a temporal holdout containing January and July 2025; the public leaderboard is used as a separate but adaptively queried 2026 signal. The mixture exponent and several late architecture choices were selected with leaderboard feedback, whereas the reconstruction rule was derived without it and only confirmed afterwards. We therefore treat leaderboard gains as descriptive evidence, not as an independent test. This distinction matters because repeated evaluation can overfit a public leaderboard [4].

All code, the complete experiment log with both positive and negative results, and the diagnostics that support every number in this paper are available at `https://github.com/ugo-roccamatisi/prc-data-challenge-2026`.

2 Data and task

The challenge provides 4,167,797 movements, arrivals and departures, at EDDF, EDDM, EGLL, EHAM, LEBL, LEMD, LFPG, LIRF, LSZH and LTFM for the whole of 2025, and the same fields for January and July 2026 as the ranking set. Each movement record carries the reporting airport's own timestamps: the movement time (takeoff for a departure, landing for an arrival), the block time (off-block or in-block), the scheduled time, and the stand and runway used. Movement records are left-joined onto the Network Manager flight list, which contributes the flight plan and flown trajectory timestamps, in particular the estimated off-block time and the actual off-block time of the flown trajectory, hereafter EOBT and AOBT.

In the ranking file, only two fields are blanked, and only for departures: the block time and the taxi time. Everything else remains visible, including the takeoff time, the scheduled time, the stand, the runway, every Network Manager field, and all arrival records with their in-block times and taxi-in durations. This asymmetry is consequential and we return to it in Section 6.

Two structural properties of the join matter throughout. First, about 1.5 % of departures are never matched to a Network Manager record, in which case every flight-side column is null. Second, for matched flights, the airport off-block and the Network Manager off-block do not

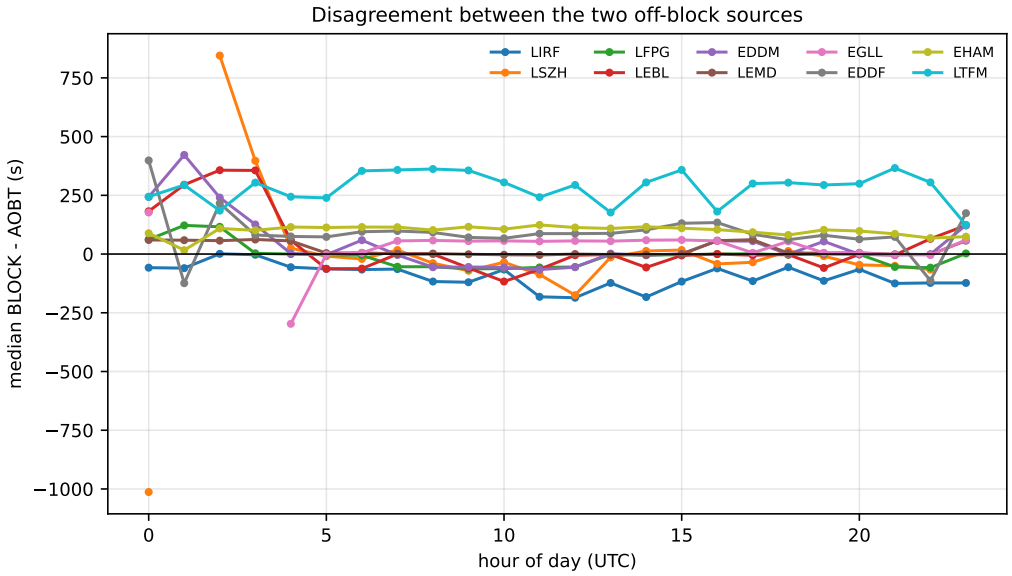


Figure 1: Hourly median of the difference between the airport block time and the Network Manager actual off-block, per airport, 2025. The curves do not scatter around a common zero: the two sources measure different events, and the per-airport offset reaches several minutes.

measure the same event. Taking the naive estimator $\widehat{y} = \text{MVT} - \text{AOBT}$ gives a root mean squared error of 380.8 s over the whole of 2025, with a per-airport hourly median bias ranging from −118 s at LIRF to 296 s at LTFM. This is not noise around a common definition; it is two definitions (Figure 1). Our models therefore predict the residual of the target over that naive estimator rather than the target itself.

3 Anatomy of the target

3.1 Schedule substitution

Selecting departures for which the recorded block time equals the scheduled time to within 10 s isolates 4.80 % of departures. On those rows the recorded taxi time is, to within a median of 3 s, exactly MVT − SCHED: the reported quantity is the sum of the departure delay and the real taxi, not the taxi. The median recorded value is 1009 s against 910 s for the rest of the population, so the contamination is invisible in a summary statistic (Figure 3), but these rows carry 29 % of the total squared error of a naive regression.

The organisers explicitly document that they did not attempt to reconcile inconsistencies between the movement and flight sources, which is consistent with what we observe: the substitution is an artefact of the reporting chain, left in place by design.

3.2 An exactly reconstructible subfamily

Among the substituted rows, the extreme tail is the one that matters for a squared-error metric. In 2025 there are 76 departures with a recorded taxi time above 20,000 s. Of these, 58 satisfy $|y - (\text{MVT} - \text{SCHED})| < 120$ s, and 72 are at LIRF. On those rows the quantity $y - (\text{MVT} - \text{SCHED})$ has a median of 1 s and an interquartile range of 6 s.

Since both the takeoff time and the scheduled time remain visible in the ranking file, these labels are recomputable rather than predictable. The difficulty is identification, not reconstruction. A threshold on MVT − SCHED alone is a poor discriminator: at 20,000 s it selects 245 candidates for 57 true positives, and applying the substitution to all of them costs more squared

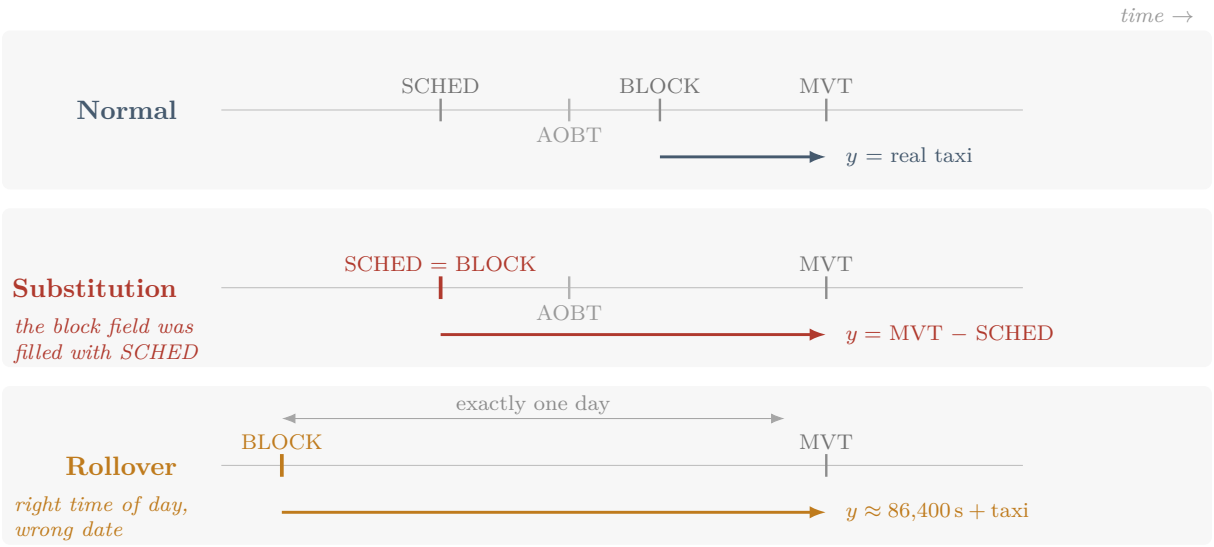


Figure 2: The three regimes of the target, drawn on a common time axis with the takeoff aligned. In a normal record the reported taxi time spans the airport block time to takeoff. Under schedule substitution the block field carries the scheduled time, so the reported value is the departure delay plus the taxi, and is exactly recomputable from two columns that remain visible at evaluation time. Under date rollover the block field carries the correct time of day with the previous day's date, which inflates the value by one day and leaves no visible trace.

error than it recovers. The discriminating conjunction turns out to involve the join structure rather than the timestamps: requiring in addition that the movement be unmatched to any Network Manager record and that the aircraft type field be empty raises precision from 0.23 to 0.85 at the same threshold.

Evaluated against our own model's predictions on a January and July 2025 holdout, hard substitution at a 10,800 s threshold restricted to LIRF reduced total root mean squared error from 472.74 s to 371.23 s. The 2026 ranking file contains 74 candidate rows, all at LIRF, against 24 in the equivalent two-month window of 2025, so the defect transfers. Measured on the leaderboard, the rule is worth 7.95 s (Table 2).

We stress that the derivation used no information from the evaluation server. It is reproducible from the 2025 training data alone; the leaderboard only confirmed it afterwards.

3.3 A family that is explicable but lacks a usable identifier

The remaining 18 aberrant labels of 2025 follow a different mechanism. Fourteen of them lie between 87,168 s and 88,392 s, that is 86,400 s plus an ordinary taxi. Inspection of the raw records shows the recorded block time carrying the correct time of day and the previous day's date: for instance a takeoff at 12:14:59 with a block time stamped at 11:55:01 the day before. The real taxi is a perfectly normal twenty minutes.

This explains the values but did not yield a usable predictor in our tests. The quantity $y - 86{,}400\,\text{s}$ matches a planned delta on one row out of eighteen, and MVT − SCHED on these rows ranges from 939 s to 93,535 s, so no threshold isolates them. The best candidate filter we could construct returns two rows in the 2026 ranking file where three or four are expected. We therefore make no correction for this family and record it as an untreated residual component. This is an empirical statement about the provided fields and tested signatures, not a proof that identification is impossible.

The contrast between the two families is the methodological point of this section. Both are reporting defects of comparable magnitude and comparable rarity. One is fully recoverable

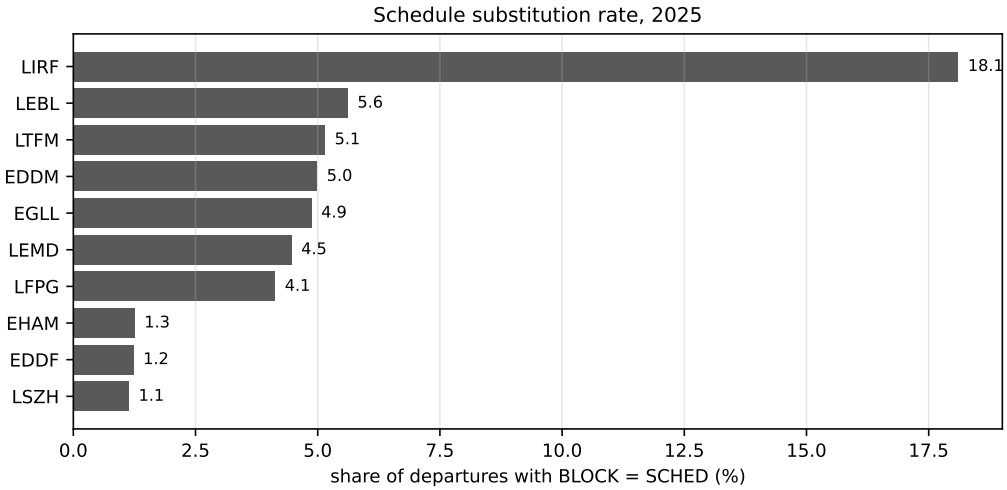


Figure 3: Share of departures whose recorded block time equals the scheduled time to within 10 s, per airport, 2025. On those rows the reported taxi time is the departure delay plus the real taxi.

Table 1: Decomposition of the validation mean squared error, January and July 2025, final configuration.

Population	Rows	Share of MSE
Labels above 20,000 s	26	58 %
Movements unmatched to NM data	5366	66 % (incl. above)
Clean lane	344,201	42 %
Worst 0.01 % of the clean lane	34	11 % of clean MSE
Worst 1 % of the clean lane	3442	51 % of clean MSE

because the defect substitutes one visible field for another; the other is not, because the defect corrupts the date of a field that is itself hidden at evaluation time.

3.4 Concentration of the error

Figure 4 and Table 1 give the decomposition of the validation mean squared error of our final model. Twenty-six rows out of 344,227, that is 0.008 % of the population, carry 58 % of it. Within the remaining clean lane, the error is well behaved: median absolute error 100 s, ninetieth percentile 306 s, ninety-ninth percentile 825 s.

A practical consequence deserves emphasis. A single defective record of the rollover family contributes roughly 21,500 to the mean squared error of the full validation set. With about twelve such events a year, a two-month evaluation window is a Poisson draw, and the public ranking of any two teams separated by a few seconds is partly decided by it.

4 Method

4.1 Regime mixture

Because the target mixes regimes, a single conditional-mean regressor is badly misspecified: we measured 472.8 s for a single model against 379.1 s for a mixture on an identical feature set. The model we use throughout predicts

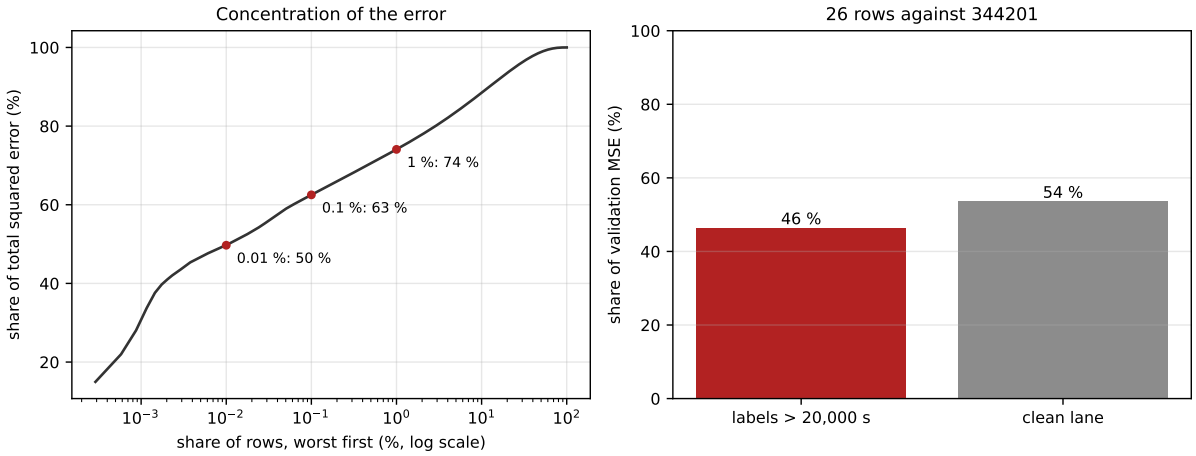


Figure 4: Left: cumulative share of the *total* validation squared error as a function of the share of rows, worst first, logarithmic abscissa; the per-lane percentages quoted in the text and in Table 1 are computed within the clean lane instead. Right: approximate split of the total between the aberrant labels and the clean lane; the exact values given in Table 1 are from the final configuration.

$$\widehat{y} = p^{\gamma}\,(\mathrm{MVT} - \mathrm{SCHED}) + (1 - p^{\gamma})\,\mu, \qquad \gamma = 0.75, \tag{1}$$

where p is a gradient-boosted classifier estimating the probability that the off-block was substituted and μ is a regressor for the normal regime. The exponent γ was selected on the leaderboard early in the work, over the grid $\{0.5, 0.6, 0.75, 1.0\}$; the public and holdout curves were parallel with a constant offset of about 32 s (Figure 5), which is our first evidence that holdout gains transfer for properties of this kind. Because γ was selected with public feedback, its leaderboard gain is descriptive rather than an unbiased out-of-sample estimate.

The normal-regime regressor predicts the residual over the Network Manager off-block, $y - (\mathrm{MVT} - \mathrm{AOBT})$, for the 98.5 % of flights where AOBT is available, and falls back to a second regressor trained on the excess over a stand-runway unimpeded time otherwise.

4.2 Features

The feature set combines Network Manager time deltas, airport, stand, runway, operator and aircraft-type categoricals, out-of-fold target encodings of the substitution rate, arrival-context features computed from the fully observed arrivals of the evaluation period, stand-linking features derived from the previous arrival at the same stand, and unimpeded times per stand-runway pair in the style of the Performance Review Unit's methodology. To these we add the posterior sequence features of Section 6.

4.3 Architecture

The combined architecture block accounts for 5.05 s of leaderboard improvement in the sequential experiments of Table 2. It comprises, first, one regressor per airport, with a fallback for airports below a size threshold. Second, and necessarily jointly, a single global corruption classifier shared by all airports: a classifier dedicated to LIRF sees that airport's fifty corrupted labels at a far higher rate than a global classifier that dilutes them in 1.7 million rows, and overfits them. Splitting the two stages recovers the clean-lane benefit without the tail penalty. Third, an ensemble of LightGBM and CatBoost with fixed weights (0.75 on CatBoost), selected from a sweep that requires no retraining. On this problem CatBoost alone outperforms LightGBM

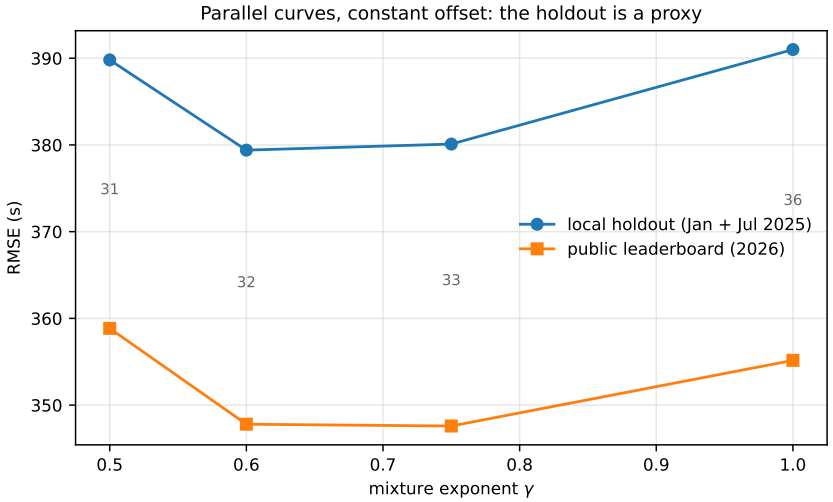


Figure 5: Holdout and public root mean squared error as a function of the mixture exponent γ. The two curves are parallel with an offset of about 32 s, which is the basis for treating the holdout as a proxy for physical properties of the model. The same reasoning fails for properties of the reporting system, as Section 7 shows.

Table 2: Public leaderboard progression of the final pipeline. Each row adds one change to the row above.

Submission	Change	Public RMSE (s)
v19	Merged baseline pipeline	338.37
v21	Label reconstruction rule (Section 3.2)	330.42
v22	Posterior sequence features (Section 6)	326.80
v23	Double capacity, lean classifier, three-seed average	325.15
v25	Per-airport regressors, global classifier, CatBoost	320.10
v26	CatBoost weight 0.75	**319.21**

alone, 357.38 s against 362.89 s, which we attribute to its ordered target statistics on the high-cardinality stand and operator variables.

These changes were introduced and evaluated sequentially rather than in a full factorial design. The measured gain therefore supports the combined block, but does not identify a context-free causal contribution for every component.

Table 2 traces the resulting progression on the public leaderboard.

5 How much of the error floor is a data floor?

Table 3 gives the clean-lane root mean squared error per airport. Eight of the ten airports sit between 174 s and 267 s, with a median absolute error around 100 s. The literature's 75 s to 90 s on cleanly measured physical taxiing is roughly a factor of two below our result here. The disagreement between the two off-block sources has a per-airport standard deviation between 250 s and 500 s, which is consistent with a large data-construction contribution to our residual. This comparison does not show that the whole remaining gap is irreducible or exclude improvements from models we did not test.

Two airports deserve comment. LFPG appears to be a problem airport at 569 s, and is not: two rows of the rollover family account for roughly three quarters of its squared error, and the clean-lane value of 254 s is unremarkable. We initially misdiagnosed this as a de-icing

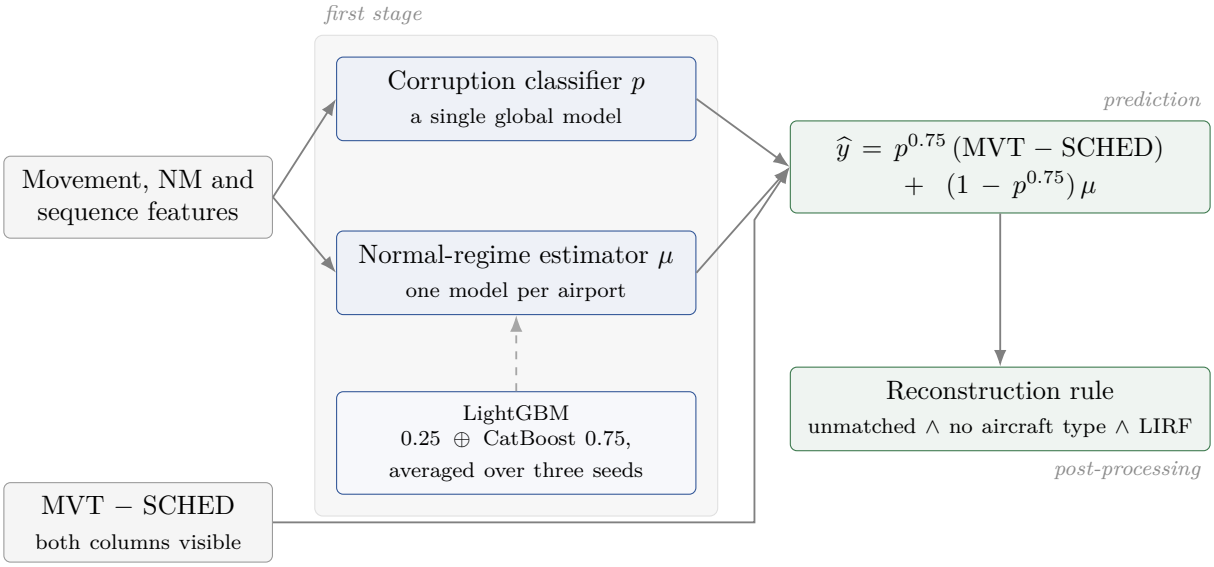


Figure 6: The prediction pipeline. The corruption classifier is deliberately global while the normal-regime estimator is fitted per airport: a classifier dedicated to LIRF sees that airport's corrupted labels at a far higher rate than a global one, and overfits them. Separating the two stages recovers the clean-lane benefit of splitting without the penalty on the corrupted tail. The reconstruction rule is applied last, on a narrowly identified subset.

effect, on the strength of a January standard deviation of 937 s against 350 s for the rest of the year, and spent effort building degraded-day detectors before decomposing the error properly. LIRF, conversely, is genuinely harder: its clean-lane error remains above 550 s, and structured estimators do no better there than a constant in our experiments, which is consistent with target noise below the aberrance threshold rather than merely tail contamination.

The aberrant rows are present in every team's ranking set, so a first-order reading of the leader's score would put their clean-lane error near 200 s against our 260 s. That inference assumes they incur the same error as we do on the aberrant rows, which need not hold if they apply a reconstruction rule of their own, and we cannot verify it. What the decomposition does establish is that the artefacts dominate the absolute value of the metric while leaving substantial room in the ordinary taxi: our own clean-lane error is 260 s against a literature floor roughly half that.

6 Directionality of the congestion screening

The strongest predictor in the taxi-time literature is the state of the departure queue at pushback [1, 2]. We implemented it twice, in two independent forms: queue length and recent throughput anchored on the aircraft's own off-block, and a demand queue counting flights whose planned pushback preceded the takeoff without having departed. Both were neutral to slightly negative, and no such feature appeared in the top fifteen by gain in any configuration.

The explanation is conditioning. Once the Network Manager off-block is observed, the model already knows the duration of the flight's own ground phase as measured by the second source; the queue that produced that duration adds nothing. Congestion is screened off by a variable that the operational forecasting problem does not have.

The screening is directional, however, and this is the result we consider most transferable beyond this dataset. The ranking file contains every movement of January and July 2026, including the departures that took off after the target flight. Building features from those, the intervals to the next takeoffs on the same runway, the forward density windows, the position of

Table 3: Clean-lane root mean squared error per airport, January and July 2025 validation, final configuration. The full column includes the aberrant labels.

Airport	Departures	Full (s)	Clean lane (s)
LIRF	26,528	836.3	557.1
EGLL	40,207	267.1	267.1
LFPG	39,577	569.2	254.1
LTFM	46,539	246.5	246.5
LEBL	28,981	218.2	218.2
LSZH	22,204	206.2	206.2
EHAM	40,942	188.8	188.8
EDDF	36,830	183.4	183.4
EDDM	27,091	179.2	179.2
LEMD	35,328	174.1	174.1

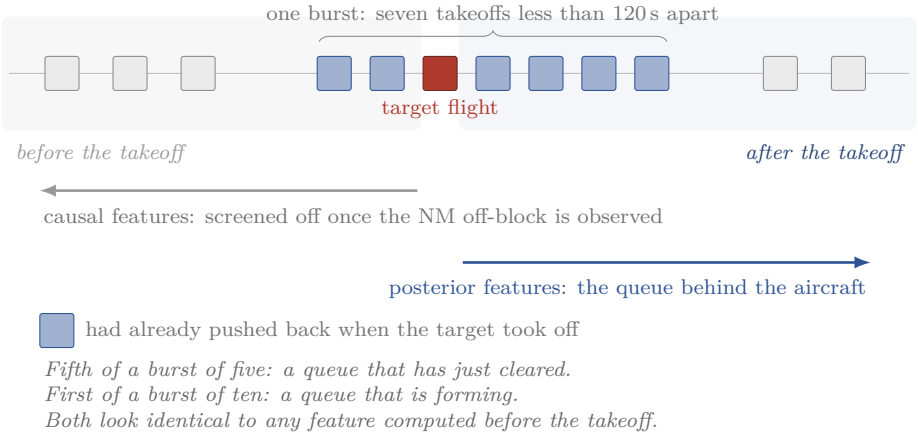


Figure 7: Why the congestion screening is directional. Each square is a takeoff on the same runway. Features computed from earlier takeoffs add nothing once the Network Manager off-block is observed, because that timestamp already encodes the queue the aircraft went through. Features computed from later takeoffs, in particular the count of subsequent departures that had already left their stand, measure the queue the aircraft was part of, and no causal feature can substitute for them.

the flight within a burst of closely spaced takeoffs and the length of that burst, the throughput normalised by the runway's monthly median, and above all the number of the next ten departures that had already pushed back at the moment the target took off, yields 3.62 s on the leaderboard and improves eight airports out of ten.

That last feature is the queue behind the aircraft, and it is measurable only by looking forward. No causal feature can distinguish a flight in fifth position of a burst of five, which leaves a queue that has just cleared, from a flight in first position of a burst of ten, which enters a queue that is forming.

These features have no operational meaning. A controller does not know future takeoffs, and a model using them cannot be deployed for departure metering. They are legitimate here because the challenge is a retrospective reconstruction task: the quantity to be recovered lies in the past, and the evaluation set is a complete two-month extract rather than a stream. We believe this distinction is worth making explicit in challenge design, since it silently changes the admissible feature space and therefore what a reported score means.

Table 4: Rejected hypotheses. Effects are on the validation set unless stated otherwise.

Hypothesis	Outcome
Single regressor instead of the regime mixture	Costs 94 s: the mixture structure is necessary
Queue and throughput features anchored on pushback	Neutral, twice, in two independent implementations
Demand queue anchored on scheduled and estimated off-block	−1.6 s
Stand-neighbourhood, airport-level and arrival-context features (34 features)	−3.0 s
Degraded-day detection from arrival taxi-in relative to its hourly norm	No effect: a de-icing aircraft delays itself, not the arrivals
Meteorological features (temperature, dew-point spread, precipitation codes)	Degraded July: temperature acts as a season surrogate under month-holdout validation
Removal of seasonal proxies	Gains 5.2 s on holdout, close to the observed three-seed range; another team reported the same change at 43 s on cross-validation and −7.6 s on the leaderboard
Upper clipping of predictions	Degrades at every threshold tested: the extreme labels are real
Expectation correction for the rollover family, calibrated on 2025 rates	19.8 s on holdout with a positive month-transfer control, −2.45 s on the leaderboard
Open ADS-B derived off-block times as an external source	Ground coverage between 0.2 % and 16 %; first detection trails the off-block by 700 s to 1300 s

7 Negative results and measured variability

7.1 What did not work

Table 4 lists the substantive ideas we tested and rejected. We report them because the effort they represent is larger than the effort behind the positive results, and because several are ideas a reader would otherwise be likely to try.

The rollover correction deserves a note of its own. It was validated by a month-transfer protocol within 2025, improved both validation months, and still lost on the leaderboard. The subset exists in 2026 with a comparable size, but its composition changed: fewer defects, more normal flights. With about twelve events a year, a two-month window does not estimate a rate. The general lesson is that modelling a data defect from its historical frequency assumes the producing system is statistically stationary at the evaluation scale, and that assumption is not testable within the data.

7.2 How large is a real effect?

Late in the work we measured the seed-to-seed spread of a single fixed configuration: three seeds gave 366.43 s, 366.17 s and 363.52 s, a range of 2.9 s. Averaging seeds with equal weights improved on the best individual seed, consistent with the observation, also reported by other participants, that equal weights hold where fitted blend weights do not. Random training and data partitioning are known to produce non-negligible performance variation in machine-learning comparisons [5].

Three runs are insufficient to estimate a variance or a general noise floor. The observed range is instead a warning threshold for this configuration. It renders a substantial share of our own single-seed comparisons inconclusive, including four rejections in Table 4 whose magnitude is of the same order. We keep them with that caveat because a change below the observed spread is not refuted; it is unresolved.

Three changes both exceed the observed three-seed range and were confirmed on the leaderboard: the reconstruction rule at 7.95 s, the posterior sequence features at 3.62 s, and the combined architecture block at 5.05 s. This is stronger evidence, although the leaderboard is adaptive and therefore not an independent confirmation. We would encourage competition write-ups to report multi-seed distributions alongside ablation tables; our three-run diagnostic changed the reading of the entire table but should not be mistaken for a precise uncertainty estimate.

8 Discussion

8.1 Implications for operational data

One major term in our residual is the disagreement between two off-block definitions maintained by two systems. A model restricted to the current columns cannot deterministically resolve which timestamp definition produced each label, because that provenance is hidden. For the data producers, this suggests that publishing a provenance flag alongside the block time, and a substitution indicator where the scheduled time was used as a fallback, would be worth more than any modelling effort applied downstream. The same holds for the rollover defect: a single boolean recording that a date was inferred rather than observed would convert a currently untreated component into a trivial one.

8.2 Limitations

Our evaluation rests on two months of 2025 as a proxy for two months of 2026, which is the only protocol the data allow but which cannot detect non-stationarity of the kind that defeated the rollover correction. We had no access to the airport systems that produce the block times, so our characterisation of the defects is inferred from their traces rather than confirmed at the source. Finally, our absolute score is not competitive, and we cannot exclude that part of the gap to the top of the leaderboard reflects modelling choices we did not explore rather than the data contribution we describe; published repositories from better-ranked teams suggest substantially larger feature sets and substantially larger models rather than a different structural insight, but we did not verify this.

Our uncertainty analysis is also limited: the 2.9 s range comes from only three seeds of one configuration and is neither a confidence interval nor a transferable noise floor. Several late choices, including the mixture exponent and ensemble weight, were informed by repeated public leaderboard queries, so their reported leaderboard gains are descriptive and may be optimistically biased. Finally, the architecture block was evaluated sequentially rather than through a full factorial ablation.

8.3 Reproducibility

The repository provides the final command line, fixed seed list, complete version log, figure-generation script and diagnostics supporting the reported counts. The final model uses only challenge data; rejected external-data experiments are kept separately. Validation trains on the remaining months of 2025 and evaluates January and July 2025, matching the ranking months. The three-seed, per-airport LightGBM/CatBoost run takes several hours on 12 CPU cores. Python dependencies are specified by minimum version in `requirements.txt`; the challenge data are not redistributed and must be obtained under their original terms.

9 Conclusion

The taxi-out times distributed in this challenge are, for a small but decisive fraction of departures, not taxi-out times. We characterised two families of defects, showed that one is exactly reconstructible from visible columns and found no usable identifier for the other among the provided fields and tested signatures. We quantified how strongly the residual of our pipeline is associated with the coexistence of two off-block definitions. We showed that the classic congestion predictors of the literature are screened off in this setting while their time-reversed counterparts are not, a distinction that follows from the task being reconstruction rather than forecasting. Finally, a three-seed diagnostic showed that several small single-seed effects in our own log are unresolved.

The practical message is that, for our pipeline, the measurement chain is a major determinant of the reported error. Better provenance would remove a substantial ambiguity, while the gap to stronger models shows that modelling choices still matter.

Acknowledgements

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